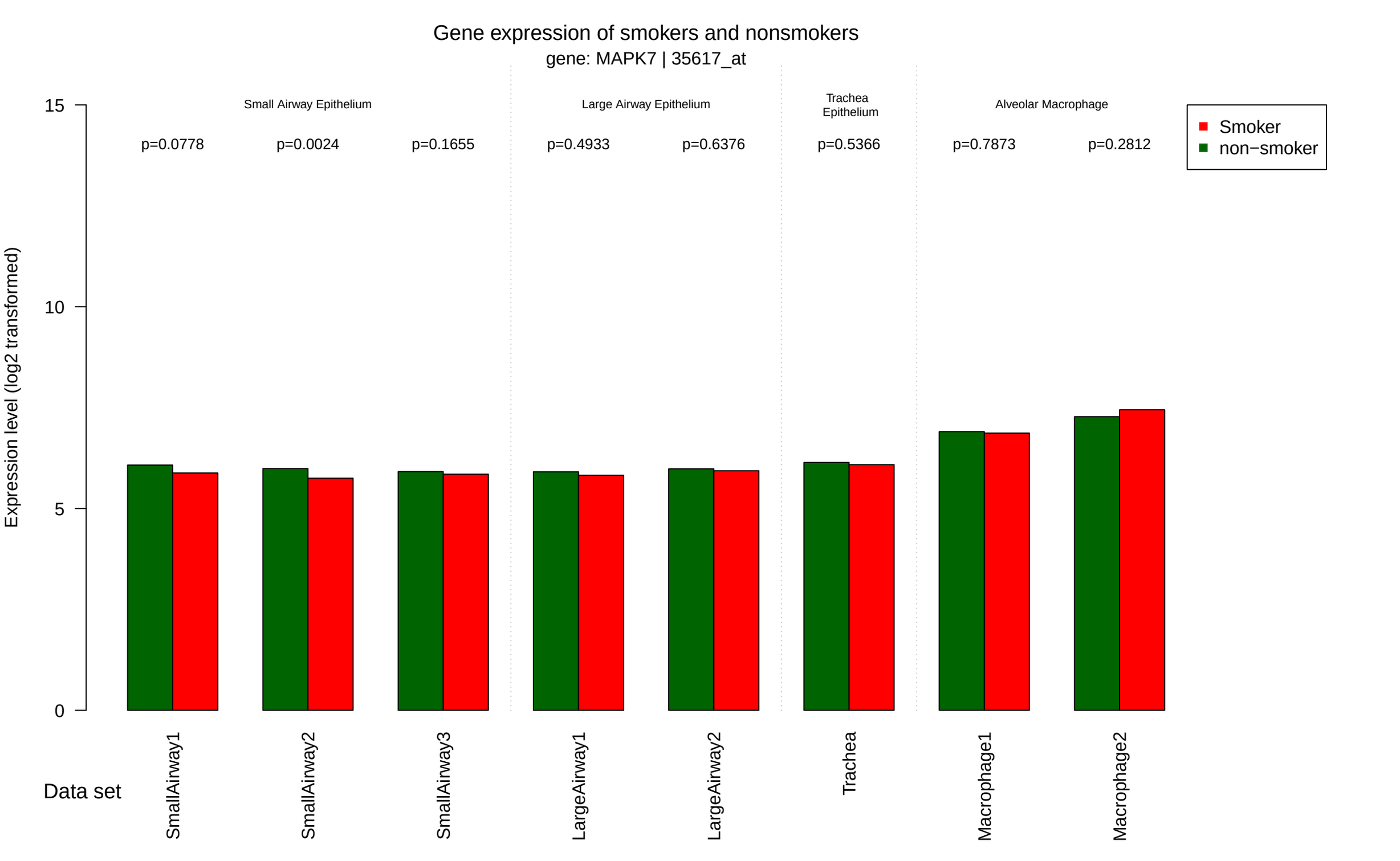
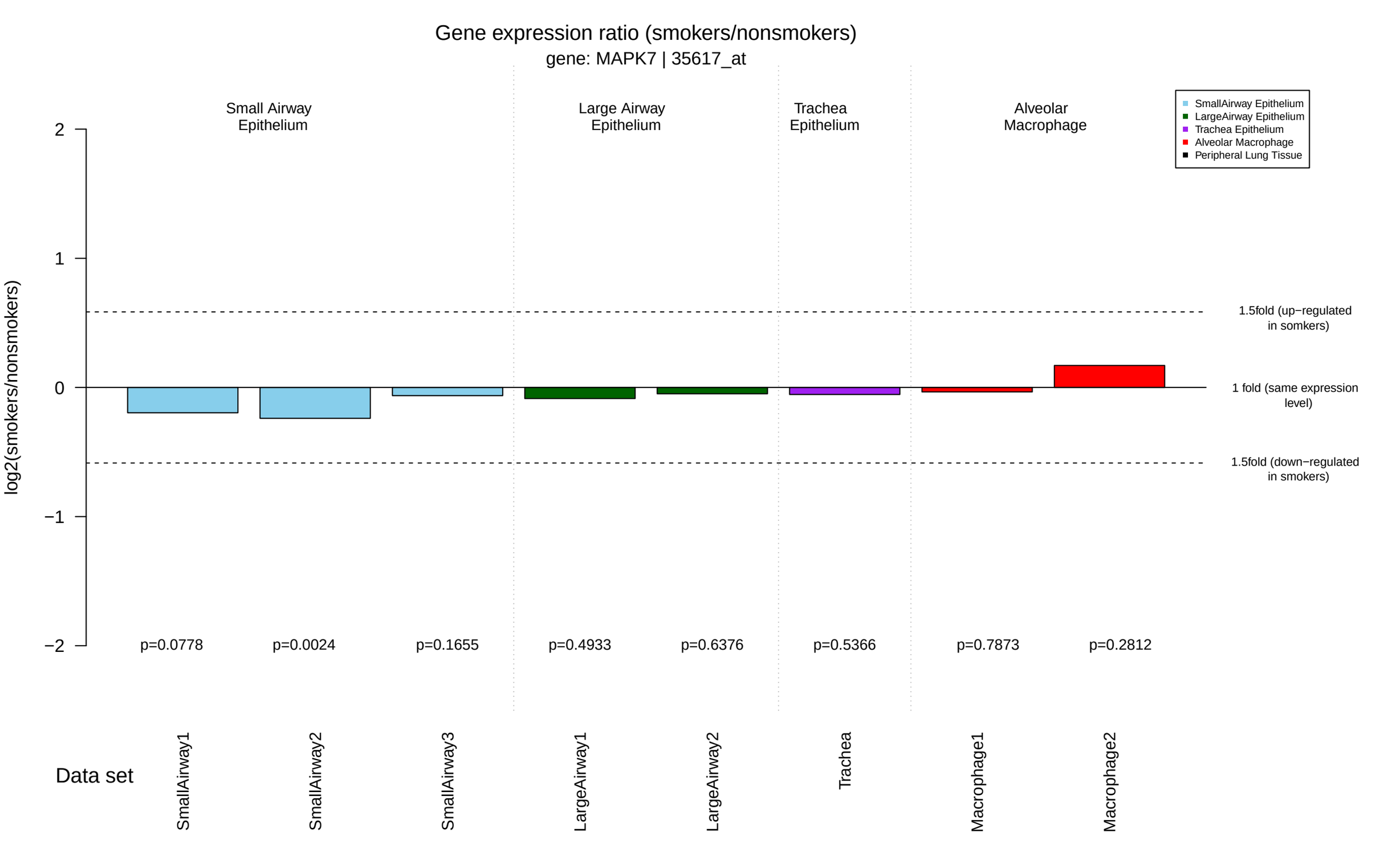
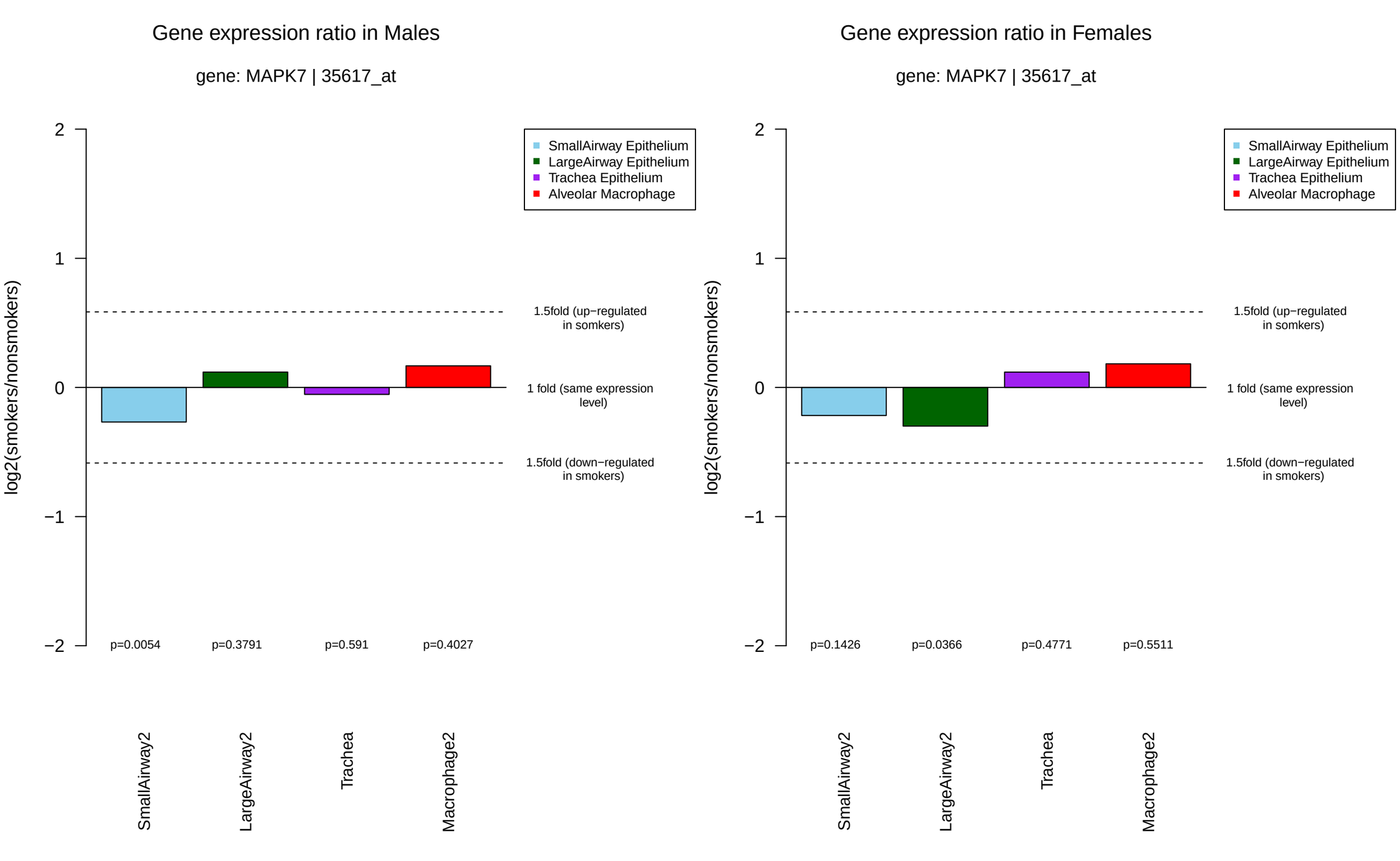
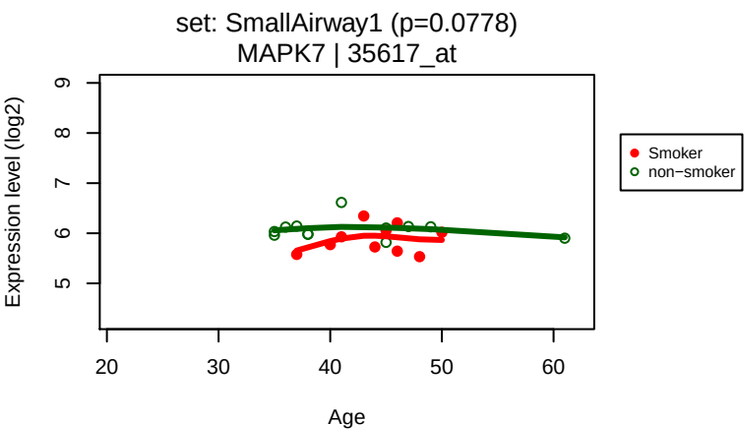
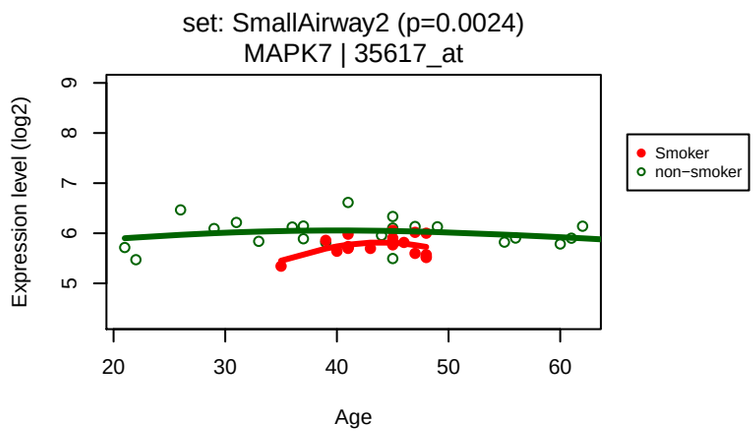
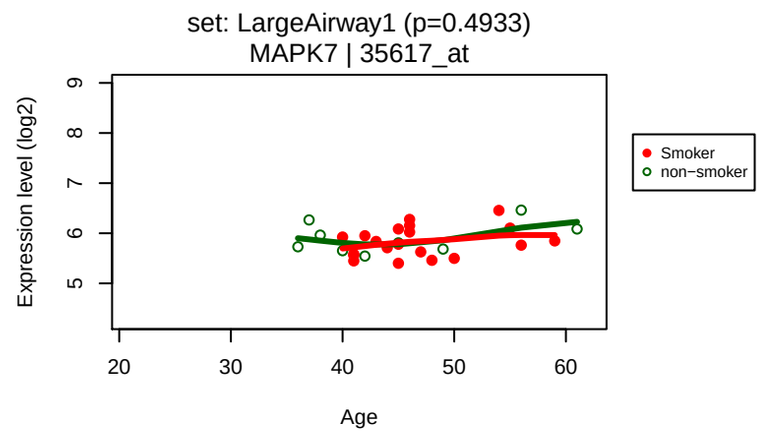

Gene expression of smokers and nonsmokers
gene: MAPK7 | 35617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0778
p=0.0024
p=0.1655
p=0.4933
p=0.6376
p=0.5366
p=0.7873
p=0.2812
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

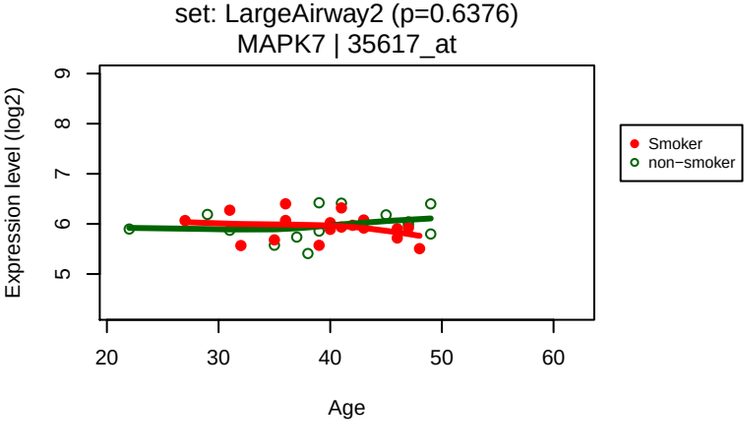

Gene expression ratio (smokers/nonsmokers)
gene: MAPK7 | 35617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0778
p=0.0024
p=0.1655
p=0.4933
p=0.6376
p=0.5366
p=0.7873
p=0.2812
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

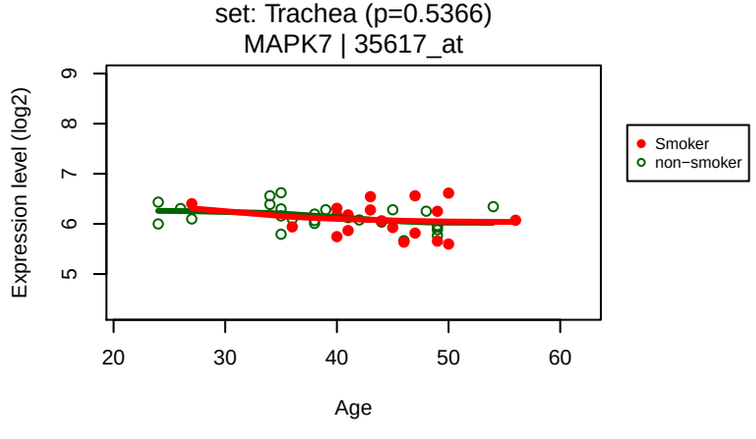

Gene expression ratio in Males
gene: MAPK7 | 35617_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.3791
p=0.591
p=0.4027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK7 | 35617_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1426
p=0.0366
p=0.4771
p=0.5511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

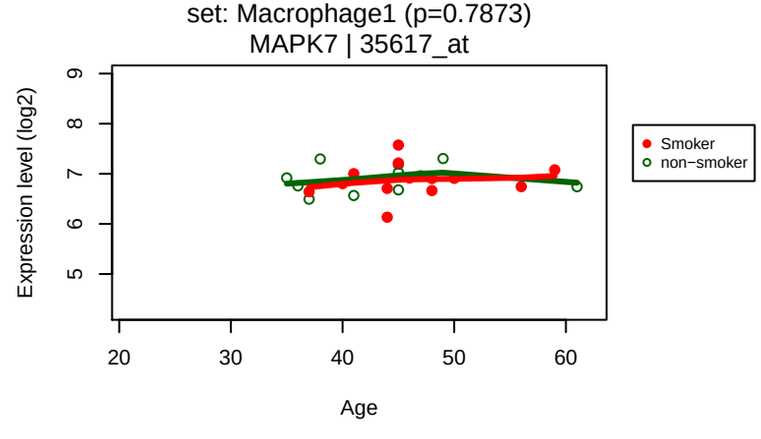

set: SmallAirway1 (p=0.0778)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker

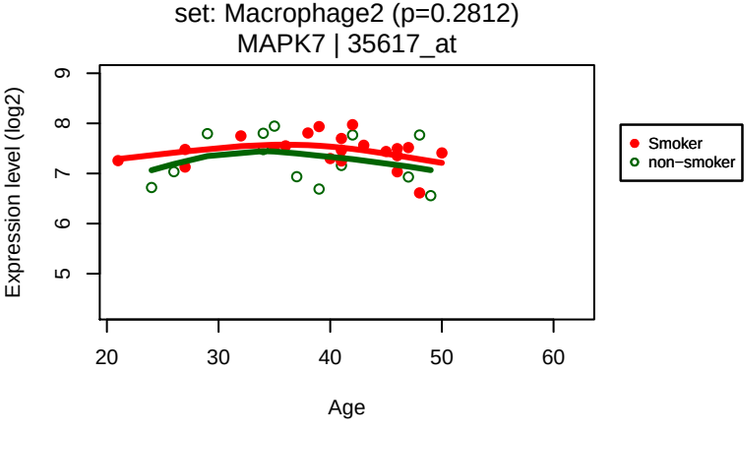

set: SmallAirway2 (p=0.0024)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker

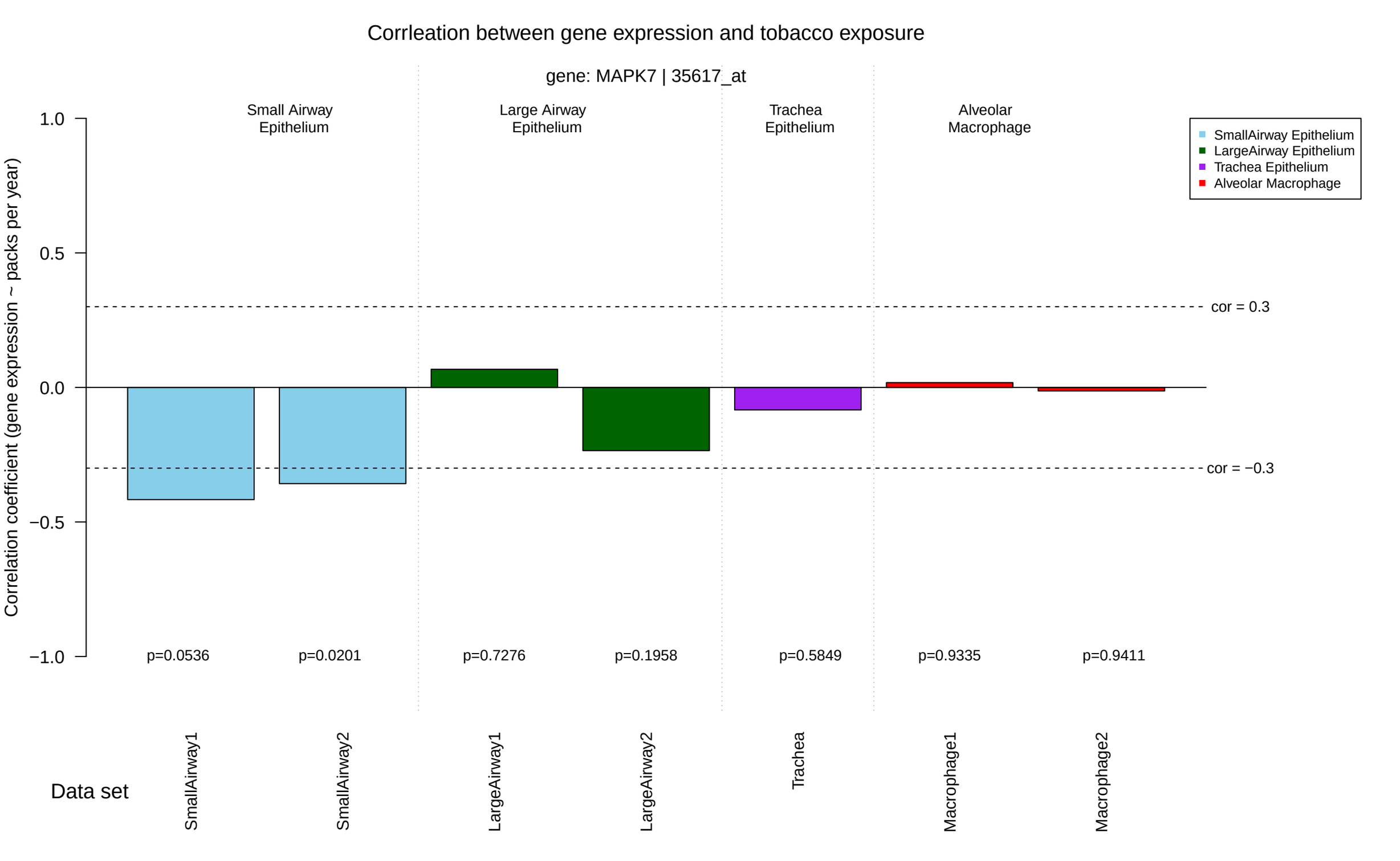

set: LargeAirway1 (p=0.4933)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6376)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5366)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7873)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2812)
MAPK7 | 35617_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK7 | 35617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0536
p=0.0201
p=0.7276
p=0.1958
p=0.5849
p=0.9335
p=0.9411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2